



Google Play Protect

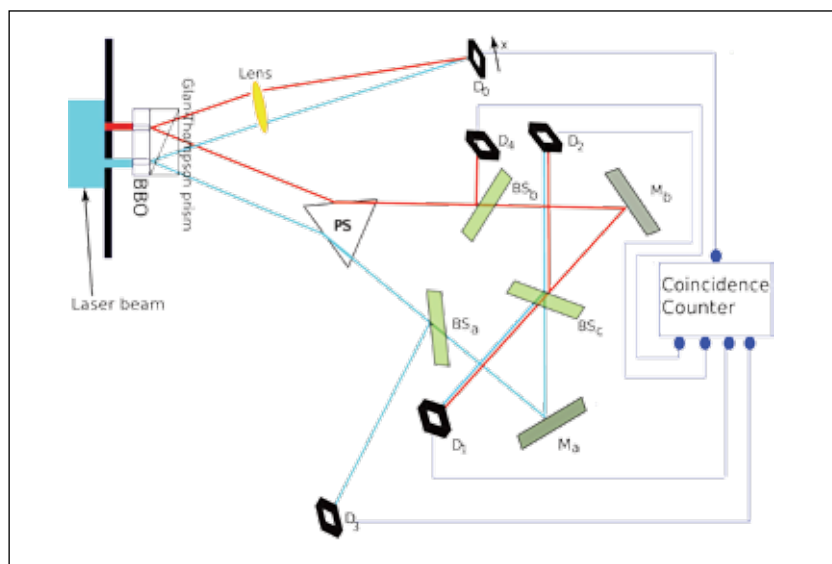
Play Protect security services to now be available to all Android users.
<http://dgit.in/playprotect>



Largest autonomous driving dataset

Nexar releases 55K street pics from 80 countries to spur autonomous driving.
<http://dgit.in/autodriveset>

CUTTING EDGE



The not-so-simple experimental setup of a quantum 'eraser'

So crazy is this thought, that Einstein himself refused to believe it was possible because it went against the laws of classical electrodynamics. He called it “spooky action at a distance” – “spooky” because it made no sense and there was obviously something wrong with it. However, over the years, we’ve come to realise that it was Einstein who was wrong.

However, there are those that disagree and maintain that Einstein was essentially right. Although the quantum coupled photons, atoms, electrons, etc., seem to allow faster than light communication, we aren’t sure how that would work. Let’s say you are trying to communicate with someone 1 light hour away, using a bunch of quantum entangled stuff exchanged between you and them earlier. Now what? Let’s say you read or write to 10 of those qubits. How would those people 1 light hour away know to read their qubits to understand that you’re communicating with them? They cannot monitor qubits, because the very act of reading a qubit changes it. They’d still have to wait for a regular bit from you (travelling at the speed of light) to tell them, “Hey, sent you a message, read qubits 1 to 10.” In that sense, Einstein was also perfectly correct in stating that even communication cannot travel faster than light,

but how do the atoms “know” when the other changes, instantly? We have no clue, really...

SECURITY

Classical cryptography is an expensive process in terms of time, skill, and even computing power. Imagine the perfect security system, in which even the act of reading the data would destroy it. Because the laws of quantum mechanics state that it is impossible to observe a quantum object without affecting it, it’s akin to someone not being able to read data without changing it. Imagine writing your password down on a piece of magic paper, which, if read, will magically change the written password to gibberish. If someone else read it, you’d know, immediately.

What this means is that information stored in a qubit is tamper-proof using the rules of physics.

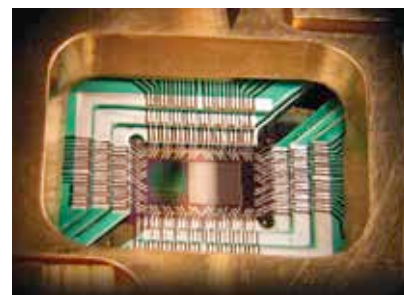
Work on quantum cryptography has only just begun, and accuracy leaves much to be desired, though the fascinating field of Quantum Error Correction has been showing some promise recently. Similarly, algorithms for quantum computation have been found to be exponentially more efficient, which started another field called Quantum Complexity Theory.

An example of this is Shor’s algorithm, which is a quantum algorithm

written for qubits, which could find prime factors of a given integer N . Normal computers can also do this, but when N is very large, regular computers would take way too long to crack this. A proper quantum computer with enough qubit crunching capabilities would do this orders of magnitude faster than a regular computer. The popular RSA public-key cryptosystem uses this type of encryption, a quantum computer running Shor’s algorithm would make short work of the RSA encryption. It’s widely believed that quantum computers will spell the death of public-key encryption.

CHALLENGES

While the math exists, the proofs exist, and the ideas exist, the technology doesn’t; not yet. However, ideas for the ‘teleportation’ of energy, and even of matter have been proposed as well, and given our history of innovation, you wouldn’t want to bet against us finding a way. Another more minor problem is that a completely quantum internet will require a complete replacement of all hardware to quantum-information compatible devices. From storage to interfacing equipment, and simple switches to processors, everything will



DWave, a quantum computing company, has rolled out models with up to 2000 qubit computing capacities

have to go in one massive upgrade cycle! It will probably be crystal-based, so say goodbye to the PCB as well. Don’t worry though, all your existing hardware is safe for decades to come still, but your grandkids might not be able to plug in and use any of your ‘antique’ devices, about 50 years from now.